Particleworks MPS method validation for oil piston cooling jets in a high-performance engine application

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Case Study: Oil Piston Cooling Jets

Competitive High Performance Engine Design → **Fast and Accurate Simulation Models**

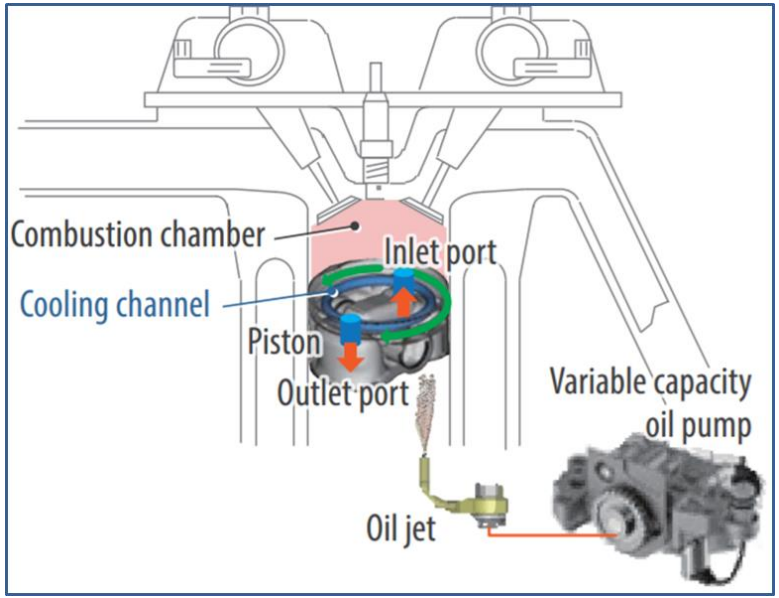
High Load on Engine Mechanical Components

↓ Mechanical Loads

↓ Thermal Loads

↓ Cooling System is needed

Case Study: Oil Piston Cooling Jets



The Oil Piston Cooling Jet is an Oil Sloshing problem about:

- **Superficial Flows**
- **Multiphase Flows**
- **Transient Flows**

Very Comoplex Model with traditional Finite Volume CFD codes

Possibility to use different simulation approach like Moving Particle Simulation

Methodology Comparison

EVO
ELECTRIFICATION SIMULATION AND AI

Finite Volume Method – Star-CCM+

- Suitable for all kind of flows and physical conditions
- Volume Mesh is needed in the entire computational domain
- Physical equations are solved on a user defined grid
- Different approach to simulate motion (from MRF to Mesh Motion)
- Geometry preprocessing needed most of the time to define Fluid Domain and Volume mesh

STAR-CCM+®

MPS method – Particleworks

- Suitable for free surface and incompressible flows
- Volume Mesh is needed in the entire computational domain
- Fluid Flow solved by particle motion equations
- Easily definition of geometry motion
- No Geometry preparation, only stl surface mesh definition and import

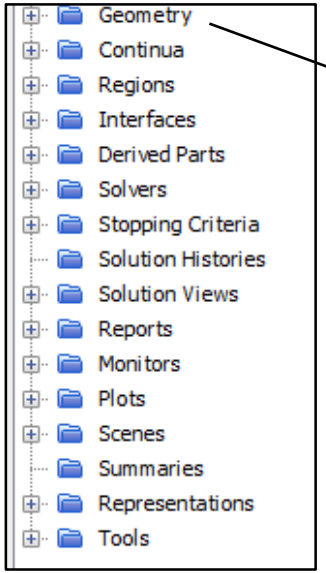
Particleworks™
Particle-based simulation software for CAE

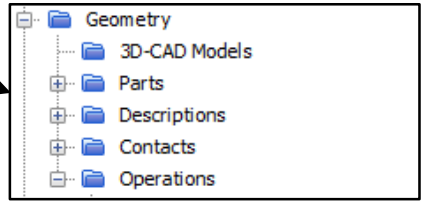
HPECOXA

Star-CCM+ Model

1. CAD Definition:

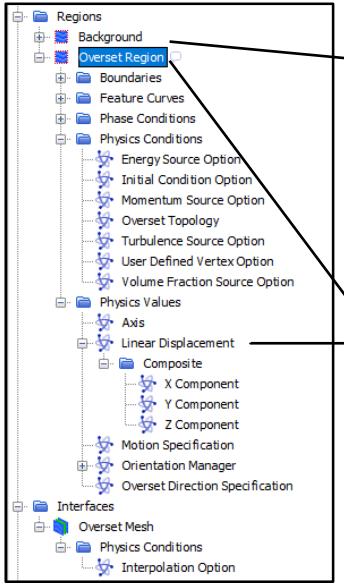
- Geometry Import
- Repair Surface and Boolean Operation
- Surface Wrapper
- Boundary Surface Definition



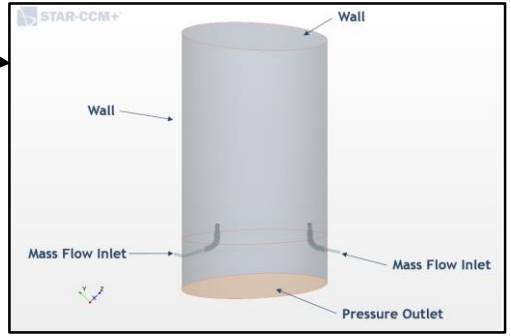


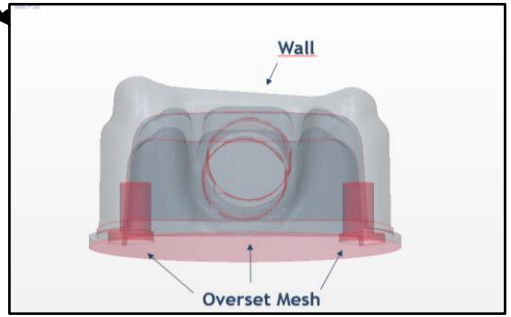
2. Motion Definition:

- Overset Mesh setup: Background and Overset regions
- Piston Linear Displacement



$$z = r\left[1 - \cos\alpha + \frac{1}{\lambda}\left(1 - \sqrt{1 - \lambda^2 \sin^2\alpha}\right)\right]$$

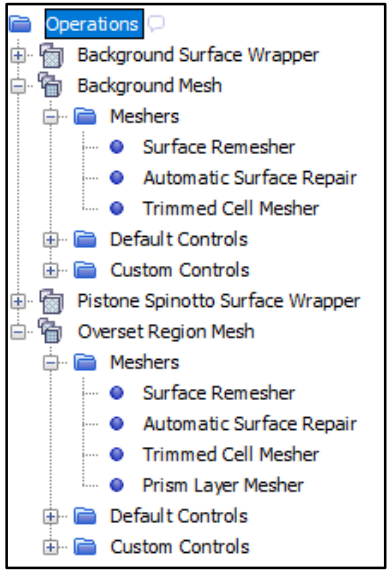


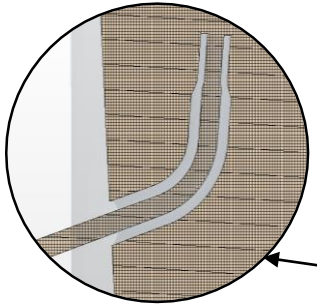


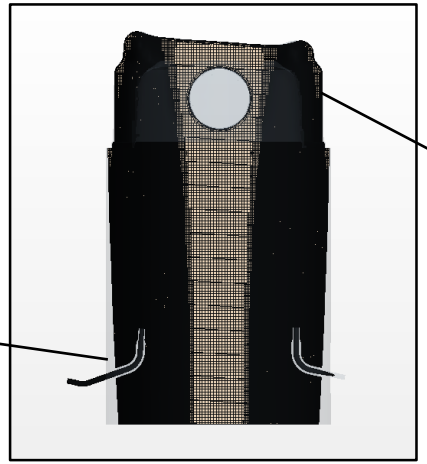
Star-CCM+ Model

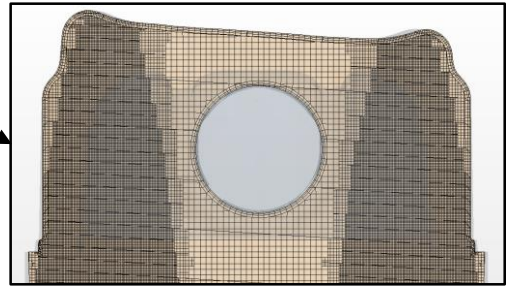
3. Mesh Definition:

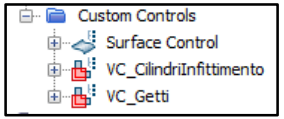
- Trimmed Mesh
- Local Refinement









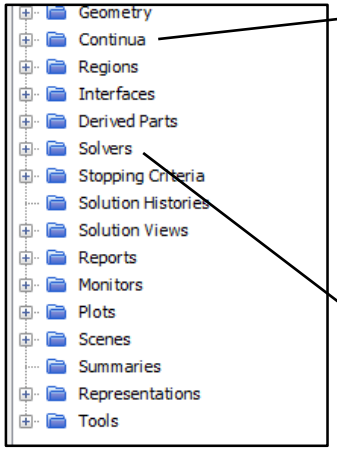


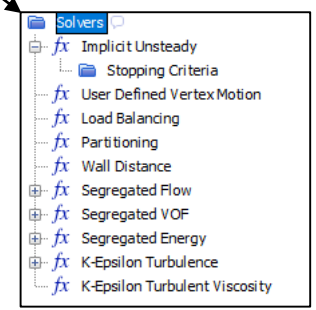
Total Volume Mesh about 15 millions cells

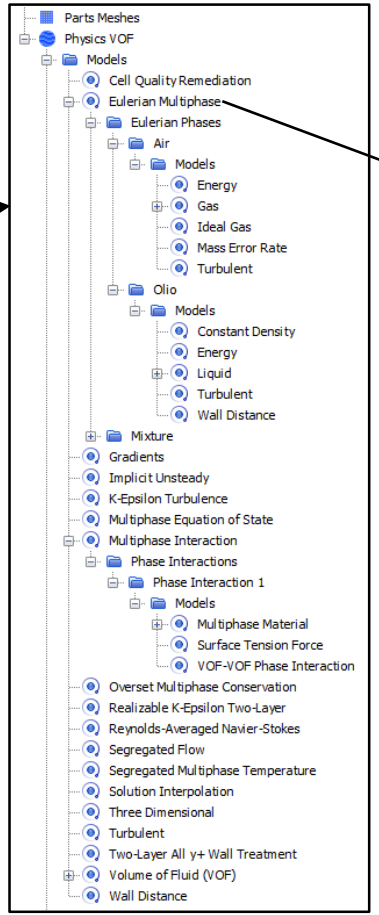
Star-CCM+ Model

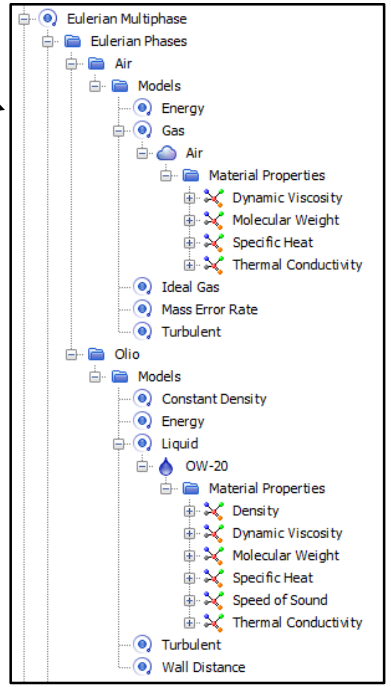
4. Solver Definition:

- Volume of Fluids Multiphase Model
- Segregated Flow Solver
- Implicit Unsteady Solver









Total Simulation Time of 0.2 s

Time Step 1.15e-5 s corresponding on 0.5 crank deg at 7250 rpm

Star-CCM+ Model

4. Results Postprocessing:

- Free Surface Flow visualization
- Heat Transfer Coefficient mapping

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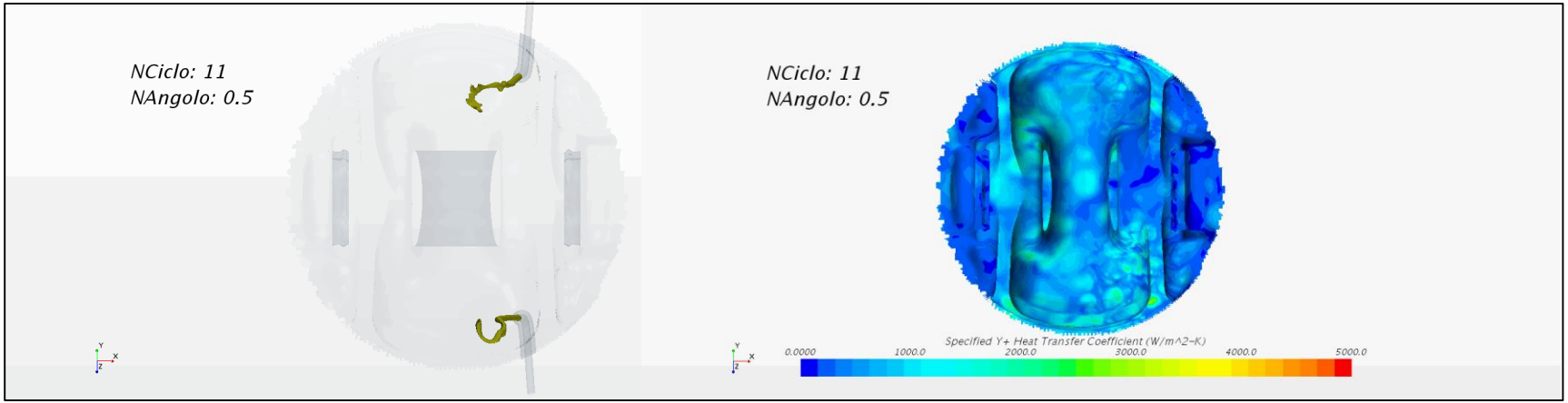
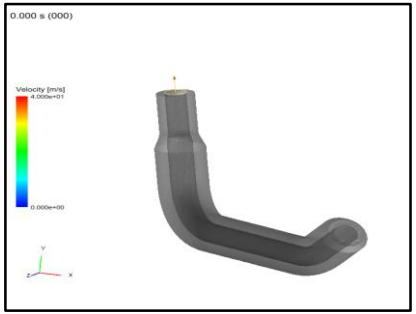
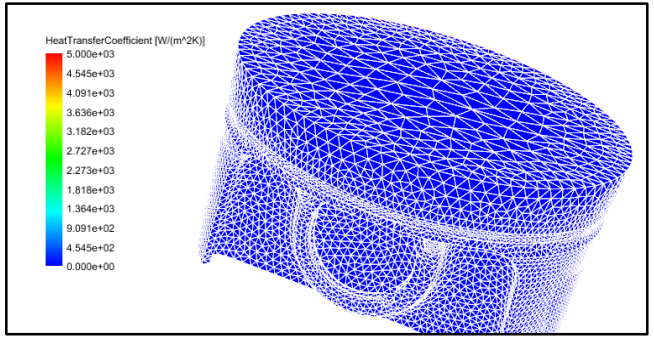


Particleworks Model

1. CAD Definition:

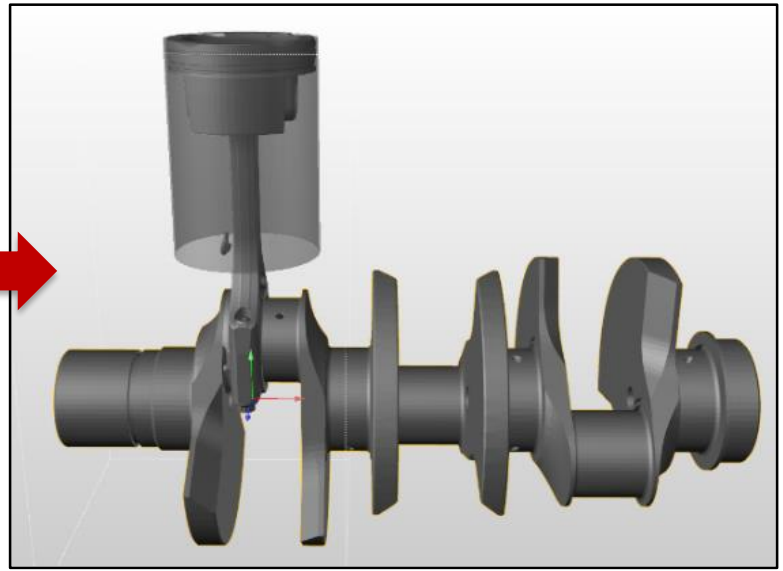
- STL definition from the CAD software
- STL import
- Parts definition





2. Motion Definition:

- Motion equation definition on moving objects



Particleworks Model

3. Physics and Solver setup:

- Particle Size and Time Step definition
- Boundary definition

Physical property manager

General | Interactions

Oil
Polygon_0
Polygon_1

Name	Oil
Density	803.2 (kg/m³)
Kinematic Viscosity	7.99e-6 (m²/s)
Surface Tension Coefficient	0.025 (N/m)
Specific Heat	1188.23 (J/kg·K)
Thermal Conductivity	0.1369 (W/m·K)
Non-Newtonian Fluid Settings	None
Stabilization	☐

pre	
Coordinate System	xyz
Particle Size	0.0002 (m)
Auto Grid Interval	☑
Flow Resistance Res.	0.5
physics	
Mass Density of Air	1 (kg/m³)
pressure	
Type	Explicit
Mode	Stabilized
Speed of Sound	234.65 (m/s)
viscosity	
Type	Explicit
surfaceTension	
Type	Potential
Only Interact Nearest DF	☐

Physics | Key Frames | Object

Inflow_Aspirazione (group: 5) ☐ Hide

Physical Property

Type: Oil

Physical Property Values

Details

Emit Mode	Volumetric flow rate
Emit Rate	0.025 (liter/s)
Upper Limit	1000 (liter)
Initial Temperature	100 (°C)
Thickness	2
Oval Spray Angle	☐
Spray Angle	0 (°)
Particle Reduction	None

Physics | Key Frames | Object

Inflow_Scarico (group: 4) ☑ Hide

Physical Property

Type: Oil

Physical Property Values

Details

Emit Mode	Volumetric flow rate
Emit Rate	0.0333 (liter/s)
Upper Limit	1000 (liter)
Initial Temperature	100 (°C)
Thickness	2
Oval Spray Angle	☐
Spray Angle	0 (°)
Particle Reduction	None

PW Wizard

Modelli_PW_2 > Oiljet_MeshLasca_3.0

Time

Finish Time	0.2 (s)
Initial dt	8.52e-7 (s)

Output Settings

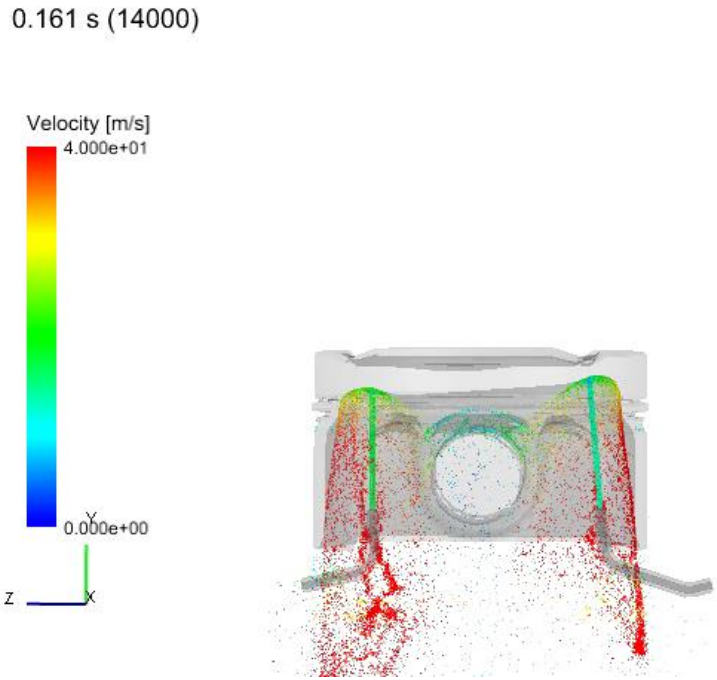
Format	v6
Precision	default
Type	simulationTime
File Output Interval Time	1.1494e-5 (s)
File Output Skip	☐
Load / Torque Type	Sampling
Load / Torque Output Interval	0.1 (s)
Restrict particles output to output regions	☐
Console Output Interval Time	0.001 (s)
Trace Log Interval Frame	1

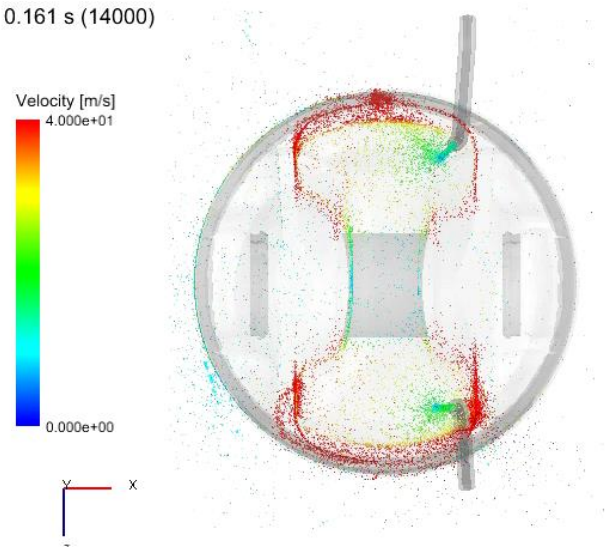
- Time Step of 8.52e-7 s due to Inertial Forces evaluation
- Particle size of 0.2 mm
- All the solver physical properties and the solver tuning are easily defined with the Particleworks Wizard and EnginSoft Best Practices.

Particleworks Results

4. Results Postprocessing:

- Free Surface Flow visualization
- Heat Transfer Coefficient mapping





Particleworks Results

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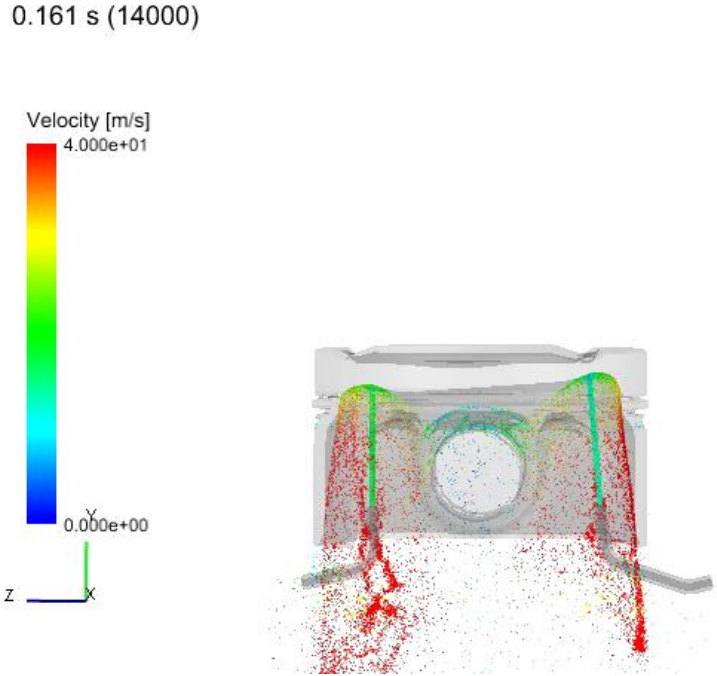
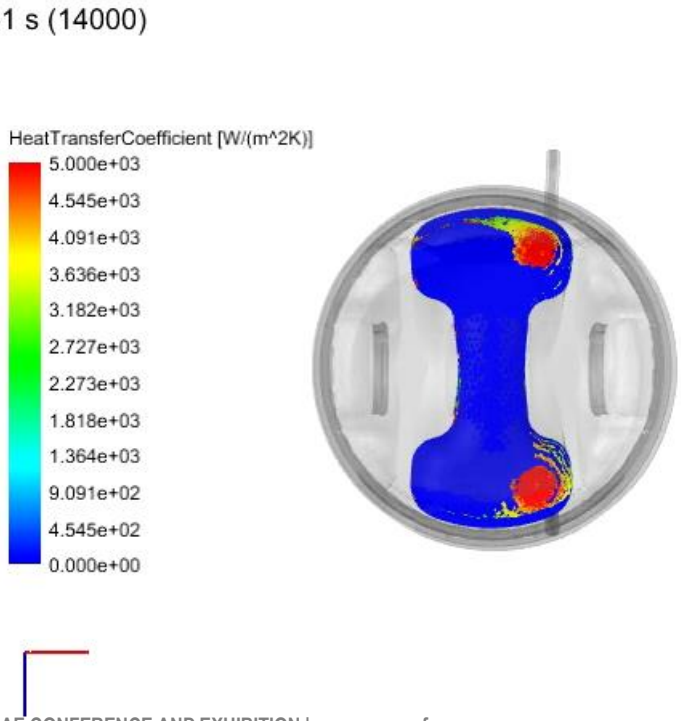
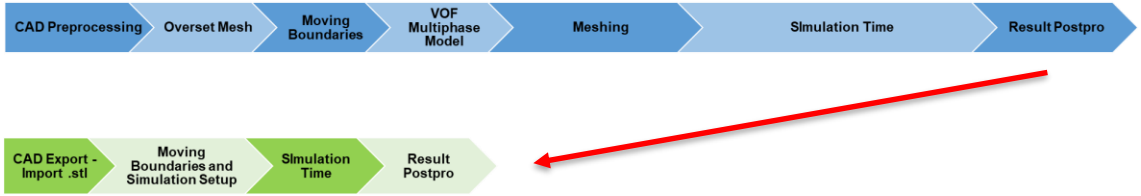


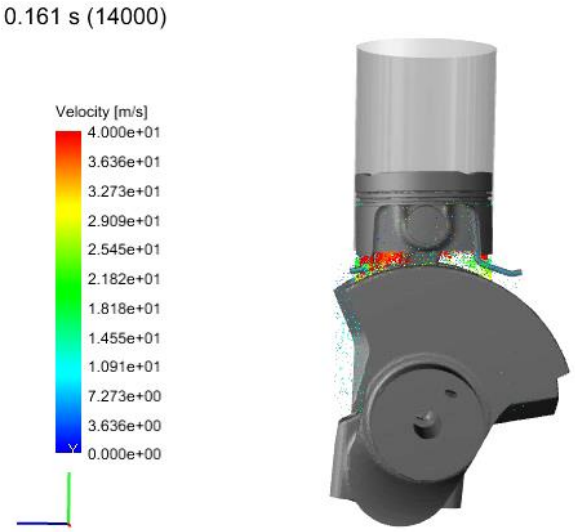


Simulation Model Comparison

The complete simulation workflow is considered to compare the total effort to produce the first result.

	Particleworks	Star-CCM+
Model Preprocessing	3	8
Simulation Setup	2	6
Simulation Time	8	96
Results Postprocessing	4	4
Total	17	114

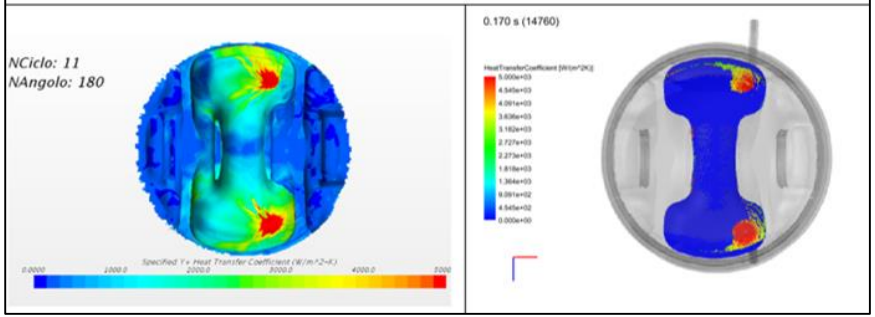




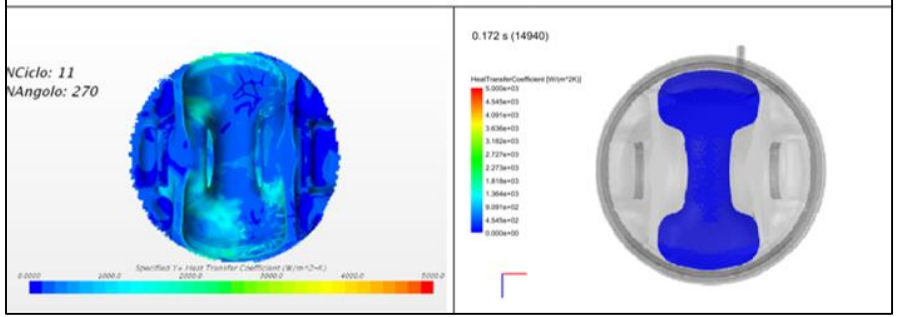
Rusults Comparison

During both simulations all the frame images about physical quantities contours was saved.
Some characteristic angle is taken into account to compare the same simulation instants in both the methodologies. Qualitative Comparison.

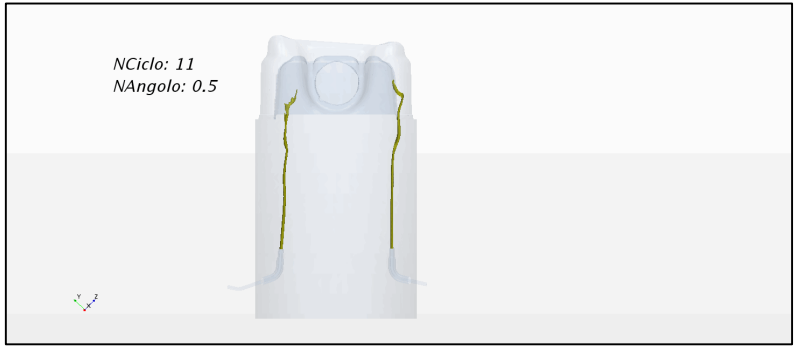
Crank Angle: 180 deg



Crank Angle: 270 deg

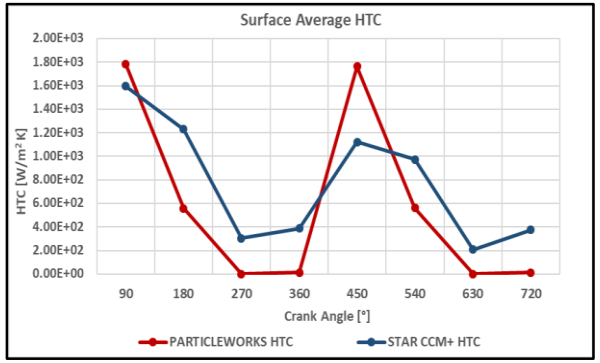


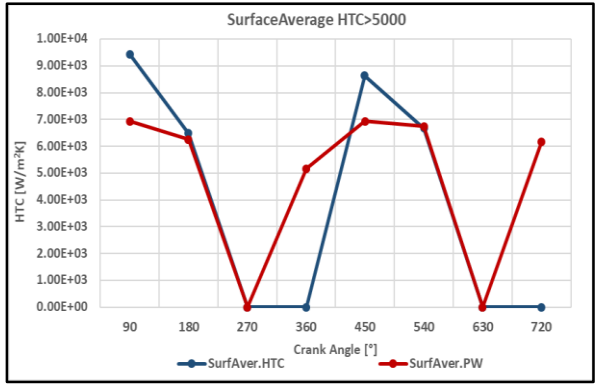
- **At 180 Crank deg the Oil Jet impact is detected and a similar contour area is visible.**
- **At 270 Crank deg both the model shows no Oil Jet Impact.**
- **In Partickleworks the HTC is evaluated only when the Oil impacts the Wall.**
- **Star-CCM+ is simulating also the HTC due to the Airflow path.**



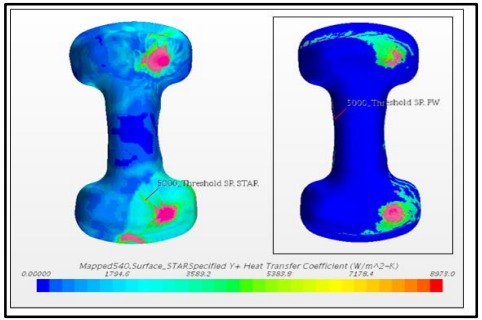
Rusults Comparison

A quantitative comparison is introduced taking into account a Surface Average HTC Value.
Only the Impact Aera is considered.



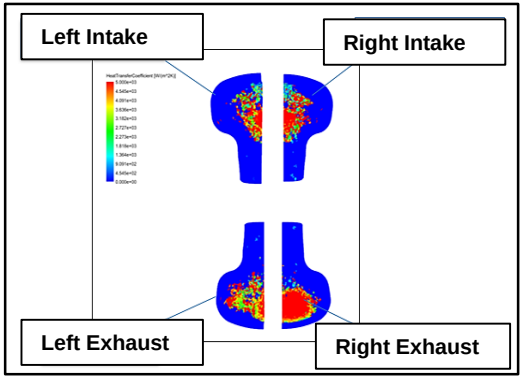


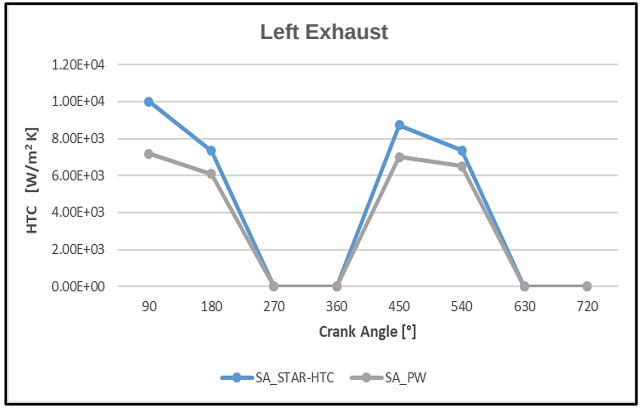
1. The complete HTC spectra is taken into account. Some difference are present but the same trend is detected.



2. Only HTC major then 5000 W/m2K are considered to filter the Airflow HTC component.

3. The Impact Area is divided in four sub-zones. The HTC trend is the same, only some difference in absolute values are present. This is due to the different HTC formulation considered by the two softwere. It can be improved by Particleworks User Scripting Routine,

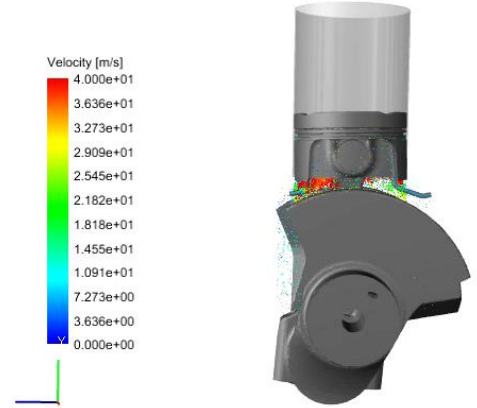




Conclusions and Next Steps

Both methodologies have some advantages:

- **Particleworks can easily consider all the moving object in the fluid domain.**
- **Particleworks can be much faster using GPUs, it can be useful to perform a lot of simulation in a predefined time (i.e. Geometrical Optimization, Targeting …).**
- **Star-CCM+ with detailed Volume of Fluids multiphase model can be very detailed but much more complex.**

Next Steps of this work will be:

1. **Piston Thermal Structural Analysis with Pistons HTC and T maps. With Particleworks a direct coupling with the FEM Ansys model is possible.**
2. **Complete Crankcase Sloshing analysis considering all the cylinders, crankshaft, Conrods, and Gerotor Suction port. This kind of analysis is possible with Particleworks. Using Star-CCM+ this model would be very complex and heavy to setup and manage.**